

Week 4 : Assignment 4

Detailed Solutions

Question 1

Let $f(x) = x^3 + 6x^2 - 3x - 5$. Select the correct options from the following:
(Select all that apply.)

- ☐ $-2 + \sqrt{5}$ will give the maximum for $f(x)$.
 - ☒ $-2 + \sqrt{5}$ will give the minimum for $f(x)$.
 - ☒ The stationary points for $f(x)$ are $-2 + \sqrt{5}$ and $-2 - \sqrt{5}$.
 - ☐ The stationary points for $f(x)$ are -4 and 0 .
-

Solution:

A stationary point occurs where the first derivative is zero.

Given

$$f(x) = x^3 + 6x^2 - 3x - 5.$$

Differentiate:

$$f'(x) = 3x^2 + 12x - 3.$$

For stationary points, $f'(x) = 0$, so

$$3x^2 + 12x - 3 = 0.$$

Dividing by 3,

$$x^2 + 4x - 1 = 0.$$

Using the quadratic formula,

$$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-1)}}{2} = \frac{-4 \pm \sqrt{20}}{2}.$$

Since $\sqrt{20} = 2\sqrt{5}$, we obtain

$$x = -2 \pm \sqrt{5}.$$

Hence the stationary points are

$$\boxed{x = -2 + \sqrt{5}, \quad x = -2 - \sqrt{5}}.$$

Now use the second derivative to classify them.

$$f''(x) = 6x + 12.$$

At $x = -2 + \sqrt{5}$,

$$f''(-2 + \sqrt{5}) = 6(-2 + \sqrt{5}) + 12 = 6\sqrt{5} > 0.$$

Since the second derivative is positive, this point is a local **minimum**.

At $x = -2 - \sqrt{5}$,

$$f''(-2 - \sqrt{5}) = -6\sqrt{5} < 0,$$

so this point gives a local **maximum**.

The correct options are: $-2 + \sqrt{5}$ gives the minimum for $f(x)$, and the stationary points are $-2 + \sqrt{5}$ and $-2 - \sqrt{5}$.

Use the following information for Questions 2 and 3.

Consider the optimization problem

$$\max_{x \in \mathbb{R}} f(x), \quad \text{where} \quad f(x) = x^4 + 7x^3 + 5x^2 - 17x + 3.$$

Let x^* be the maximizer of $f(x)$.

Question 2

What is the second order sufficient condition for x^* to be the maximizer of the function $f(x)$?

- $4x^3 + 21x^2 + 10x - 17 = 0$
- $12x^2 + 42x + 10 = 0$
- $12x^2 + 42x + 10 > 0$
- ✓ $12x^2 + 42x + 10 < 0$

Solution:

For a differentiable function, a necessary condition for an interior stationary point is

$$f'(x^*) = 0.$$

For a maximum, the second derivative must be negative:

$$\boxed{f''(x^*) < 0}.$$

First calculate the derivatives. Given

$$f(x) = x^4 + 7x^3 + 5x^2 - 17x + 3,$$

we get

$$f'(x) = 4x^3 + 21x^2 + 10x - 17.$$

Differentiating again,

$$f''(x) = 12x^2 + 42x + 10.$$

Therefore, the second-order sufficient condition for a maximum is

$$\boxed{12(x^*)^2 + 42x^* + 10 < 0}.$$

Question 3

Find the value of x^* .

- ☐ -4.48
 - ☐ 0.66
 - ☒ -1.43
 - ☐ 4.45
-

Solution:

For a stationary point, $f'(x) = 0$. We already found

$$f'(x) = 4x^3 + 21x^2 + 10x - 17.$$

Therefore, we solve

$$4x^3 + 21x^2 + 10x - 17 = 0.$$

The three roots are approximately

$$x \approx -4.48, \quad x \approx -1.43, \quad x \approx 0.66.$$

To determine which one corresponds to a maximum, use

$$f''(x) = 12x^2 + 42x + 10.$$

For $x \approx -1.43$,

$$f''(-1.43) = 12(-1.43)^2 + 42(-1.43) + 10 \approx 24.53 - 60.06 + 10 \approx -25.53 < 0.$$

Therefore, $x = -1.43$ satisfies the second-order sufficient condition for a maximum.

$$\boxed{x^* \approx -1.43}$$

Question 4

Let $f(x) = 2 \sin x$, $0 \leq x \leq 2\pi$. Select the correct options from the following:
(Select all that apply.)

- ☒ $\frac{\pi}{2}$ is the global maximum of $f(x)$.
- ☐ π is the global minimum of $f(x)$.
- ☐ $\frac{3\pi}{2}$ is the global maximum of $f(x)$.
- ☒ $\frac{3\pi}{2}$ is the global minimum of $f(x)$.

Solution:

Given $f(x) = 2 \sin x$. We know that $-1 \leq \sin x \leq 1$. Multiplying throughout by 2,

$$-2 \leq 2 \sin x \leq 2.$$

Therefore, the maximum value is $f(x) = 2$ when $\sin x = 1$. In the interval $[0, 2\pi]$,

$$\sin x = 1 \quad \text{at} \quad x = \frac{\pi}{2}.$$

Hence, $x = \frac{\pi}{2}$ is the location of the global maximum.

Similarly, the minimum occurs when $\sin x = -1$. In $[0, 2\pi]$,

$$\sin x = -1 \quad \text{at} \quad x = \frac{3\pi}{2}.$$

Hence, $x = \frac{3\pi}{2}$ is the location of the global minimum.

The correct statements are: $\frac{\pi}{2}$ is the global maximum of $f(x)$, and $\frac{3\pi}{2}$ is the global minimum of $f(x)$.

Use the following information for Questions 5, 6, 7 and 8.

Let

$$f(x) = 2x_1^2 + 3x_1x_2 + 3x_2^2 + x_1 + 3x_2.$$

Question 5

Find the gradient for $f(x)$.

- ✓ $\nabla f = \begin{bmatrix} 4x_1 + 3x_2 + 1 \\ 3x_1 + 6x_2 + 3 \end{bmatrix}$
- $\nabla f = \begin{bmatrix} 3x_1 + 6x_2 + 3 \\ 4x_1 + 3x_2 + 1 \end{bmatrix}$

- $\nabla f = \begin{bmatrix} 4x_1 + 3x_2 \\ 3x_1 + 6x_2 \end{bmatrix}$
- $\nabla f = \begin{bmatrix} 4x_2 + 3x_1 + 1 \\ 3x_2 + 6x_1 + 3 \end{bmatrix}$

Solution:

The gradient is the vector of first-order partial derivatives:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}.$$

First differentiate with respect to x_1 :

$$\frac{\partial f}{\partial x_1} = 4x_1 + 3x_2 + 1.$$

Now differentiate with respect to x_2 :

$$\frac{\partial f}{\partial x_2} = 3x_1 + 6x_2 + 3.$$

Therefore,

$$\boxed{\nabla f = \begin{bmatrix} 4x_1 + 3x_2 + 1 \\ 3x_1 + 6x_2 + 3 \end{bmatrix}}$$

Question 6

Find the stationary point for $f(x)$.

- 0.6, 0.4
- -0.6, -0.4
- ✓ 0.2, -0.6
- 0.2, 0.6

Solution:

A stationary point occurs when the gradient is zero. From Question 5,

$$\nabla f = \begin{bmatrix} 4x_1 + 3x_2 + 1 \\ 3x_1 + 6x_2 + 3 \end{bmatrix}.$$

Set the gradient equal to zero:

$$4x_1 + 3x_2 = -1 \tag{1}$$

and

$$3x_1 + 6x_2 = -3. \quad (2)$$

From equation (2),

$$x_1 + 2x_2 = -1 \quad \Rightarrow \quad x_1 = -1 - 2x_2.$$

Substitute this into equation (1):

$$4(-1 - 2x_2) + 3x_2 = -1 \quad \Rightarrow \quad -4 - 8x_2 + 3x_2 = -1.$$

Hence,

$$-5x_2 = 3 \quad \Rightarrow \quad x_2 = -\frac{3}{5} = -0.6.$$

Now,

$$x_1 = -1 - 2(-0.6) = -1 + 1.2 = 0.2.$$

Therefore, the stationary point is

$$\boxed{(0.2, -0.6)}.$$

Question 7

Find the Hessian matrix for $f(x)$.

- $\nabla^2 f = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$
 - $\nabla^2 f = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$
 - ✓ $\nabla^2 f = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}$
 - $\nabla^2 f = \begin{bmatrix} 6 & 3 \\ 3 & 4 \end{bmatrix}$
-

Solution:

The Hessian matrix contains all second-order partial derivatives:

$$\nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}.$$

From the gradient,

$$\frac{\partial f}{\partial x_1} = 4x_1 + 3x_2 + 1 \quad \Rightarrow \quad \frac{\partial^2 f}{\partial x_1^2} = 4, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 3.$$

Similarly,

$$\frac{\partial f}{\partial x_2} = 3x_1 + 6x_2 + 3 \quad \Rightarrow \quad \frac{\partial^2 f}{\partial x_2 \partial x_1} = 3, \quad \frac{\partial^2 f}{\partial x_2^2} = 6.$$

Thus,

$$\nabla^2 f = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}$$

Question 8

The stationary point obtained in Question 6 is a

- ☐ maxima
 - ☒ minima
 - ☐ saddle point
-

Solution:

From Question 7, the Hessian matrix is

$$H = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}.$$

For a 2×2 Hessian matrix $H = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, it is positive definite if $a > 0$ and $\det(H) > 0$.

Here, $a = 4 > 0$. Also,

$$\det(H) = (4)(6) - (3)(3) = 24 - 9 = 15 > 0.$$

Therefore, the Hessian is positive definite. A stationary point with a positive definite Hessian is a local minimum.

Hence, the stationary point $(0.2, -0.6)$ is a **minimum**.

Question 9

Let $f(x_1, x_2) = 4x_1^2 - 4x_1x_2 + 2x_2^2$. Select the correct options from the following:
(Select all that apply.)

- ☐ $(2, 4)$ is a stationary point of $f(x)$.

- ☒ $(0, 0)$ is a stationary point of $f(x)$.
☒ The Hessian matrix $\nabla^2 f$ is positive definite.
☐ The Hessian matrix $\nabla^2 f$ is not positive definite.
-

Solution:

We first find the gradient. Differentiate with respect to x_1 :

$$\frac{\partial f}{\partial x_1} = 8x_1 - 4x_2.$$

Differentiate with respect to x_2 :

$$\frac{\partial f}{\partial x_2} = -4x_1 + 4x_2.$$

Thus,

$$\nabla f = \begin{bmatrix} 8x_1 - 4x_2 \\ -4x_1 + 4x_2 \end{bmatrix}.$$

For a stationary point, $\nabla f = 0$. Therefore,

$$8x_1 - 4x_2 = 0 \quad \text{and} \quad -4x_1 + 4x_2 = 0.$$

From the first equation, $x_2 = 2x_1$. From the second equation, $x_2 = x_1$. Therefore,

$$x_1 = x_2 = 0.$$

Hence, $(0, 0)$ is the stationary point.

Now calculate the Hessian:

$$\nabla^2 f = \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix}.$$

The first leading principal minor is $8 > 0$. The determinant is

$$\det(H) = (8)(4) - (-4)(-4) = 32 - 16 = 16 > 0.$$

Therefore, the Hessian is positive definite. Hence, $(0, 0)$ is a local minimum.

The correct options are: $(0, 0)$ is a stationary point of $f(x)$, and $\nabla^2 f$ is positive definite.

Question 10

In an optimization problem, the function that we want to optimize is called

- Decision function
- Constraints function

- Optimal function
- ✓ Objective function

Solution:

An optimization problem generally has a function whose value we want to maximize or minimize. This function is called the **objective function**.

For example, $\max_x f(x)$ has $f(x)$ as its objective function. Similarly, $\min_x f(x)$ also has $f(x)$ as its objective function.

Objective function

Question 11

The optimization problem $\min_x f(x)$ can also be written as $\max_x f(x)$. Is this statement True or False?

- True
- ✓ False

Solution:

The statement claims

$$\min_x f(x) = \max_x f(x).$$

In general, this is not true. Minimizing a function is equivalent to maximizing its negative:

$$\min_x f(x) = -\max_x [-f(x)].$$

It is **not** generally equivalent to $\max_x f(x)$.

For example, consider $f(x) = x^2$. The minimum occurs at $x = 0$ with value $f(0) = 0$. But x^2 has no finite global maximum over $\mathbb{R}$.

Therefore, $\min_x f(x)$ and $\max_x f(x)$ are clearly not equivalent.

False

Question 12

In the gradient descent algorithm, the step size should always be the same for each iteration. Is this statement True or False?

- True

✓ False

Solution:

The gradient descent update rule is

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k),$$

where α_k is the step size or learning rate at iteration k . The important point is that α_k does not necessarily have to be the same for every iteration.

For example, we can use a constant step size, $\alpha_k = \alpha$. However, we can also use a varying step size, such as $\alpha_k = \frac{1}{k+1}$, which decreases as the number of iterations increases. This is known as a decreasing learning-rate strategy.

Therefore, the step size does **not** have to always remain the same.

False

Answer Summary

Q	Correct Answer
1	$-2 + \sqrt{5}$ gives minimum; stationary points $-2 + \sqrt{5}$, $-2 - \sqrt{5}$
2	$12x^2 + 42x + 10 < 0$
3	$x^* = -1.43$
4	$\pi/2$ global max; $3\pi/2$ global min
5	$\nabla f = [4x_1 + 3x_2 + 1, 3x_1 + 6x_2 + 3]^T$
6	$(0.2, -0.6)$
7	$\nabla^2 f = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}$
8	Minima
9	$(0, 0)$ is stationary; Hessian positive definite
10	Objective function
11	False
12	False
